Example 6.36

Find the shortest distance between the two given straight lines $\vec{r} = (2\hat{i} + 3\hat{j} + 4\hat{k}) + t(-2\hat{i} + \hat{j} - 2\hat{k})$ and $\frac{x-3}{2} = \frac{y}{-1} = \frac{z+2}{2}$.

Solution

The parametric form of vector equations of the given straight lines are

$$\vec{r} = (2\hat{i} + 3\hat{j} + 4\hat{k}) + t(-2\hat{i} + \hat{j} - 2\hat{k})$$

$$\text{and} \quad \vec{r} = (3\hat{i} - 2\hat{k}) + t(2\hat{i} - \hat{j} + 2\hat{k})$$

Comparing the given two equations with $\vec{r} = \vec{a} + t\vec{b}$, $\vec{r} = \vec{c} + s\vec{d}$

we have $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{b} = -2\hat{i} + \hat{j} - 2\hat{k}$, $\vec{c} = 3\hat{i} - 2\hat{k}$, $\vec{d} = 2\hat{i} - \hat{j} + 2\hat{k}$.

Clearly, $\vec{b}$ is a scalar multiple of $\vec{d}$, and hence the two straight lines are parallel. We know that the shortest distance between two parallel straight lines is given by $d = \frac{|(\vec{c} - \vec{a}) \times \vec{b}|}{|\vec{b}|}$.

$$\text{Now,} \quad (\vec{c} - \vec{a}) \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & -6 \\ -2 & 1 & -2 \end{vmatrix} = 12\hat{i} + 14\hat{j} - 5\hat{k}$$

$$\text{Therefore,} \quad d = \frac{|12\hat{i} + 14\hat{j} - 5\hat{k}|}{|-2\hat{i} + \hat{j} - 2\hat{k}|} = \frac{\sqrt{365}}{3}.$$

Example 6.37

Find the coordinate of the foot of the perpendicular drawn from the point $(-1, 2, 3)$ to the straight line $\vec{r} = (\hat{i} - 4\hat{j} + 3\hat{k}) + t(2\hat{i} + 3\hat{j} + \hat{k})$. Also, find the shortest distance from the given point to the straight line.

Solution

Comparing the given equation $\vec{r} = (\hat{i} - 4\hat{j} + 3\hat{k}) + t(2\hat{i} + 3\hat{j} + \hat{k})$ with $\vec{r} = \vec{a} + t\vec{b}$, we get $\vec{a} = \hat{i} - 4\hat{j} + 3\hat{k}$, and $\vec{b} = 2\hat{i} + 3\hat{j} + \hat{k}$. We denote the given point $(-1, 2, 3)$ by D, and the point $(1, -4, 3)$ on the straight line by F. If F is the foot of the perpendicular from D to the straight line, then F is of the form $(2t + 1, 3t - 4, t + 3)$ and $\overrightarrow{DF} = \overrightarrow{OF} - \overrightarrow{OD} = (2t + 2)\hat{i} + (3t - 6)\hat{j} + t\hat{k}$.

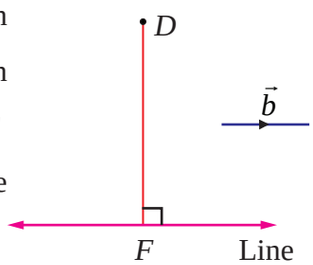


Fig. 6.23

Since $\vec{b}$ is perpendicular to $\overrightarrow{DF}$, we have

$$\vec{b} \cdot \overrightarrow{DF} = 0 \Rightarrow 2(2t + 2) + 3(3t - 6) + 1(t) = 0 \Rightarrow t = 1$$

Therefore, the coordinate of F is $(3, -1, 4)$

Now, the perpendicular distance from the given point to the given line is

$$DF = |\overrightarrow{DF}| = \sqrt{4^2 + (-3)^2 + 1^2} = \sqrt{26} \text{ units.}$$

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